

Periodical Test - II

VII - Semester

Section A

A1 (i) a)

(ii) 1. camera, wheel encoder

2. Error propagation impacts the robot's localization and path planning because by calculating the distance robot can plan the path.

A2 (i) a)

(ii)

True. If will not able to plan the path and navigate to correct place.

A3 (i) a)

(ii)

1) a)

2) Feature extraction allows robot to convert raw LIDAR and camera data into useful technique such as edges of objects.

3)

If the feature extractor is skipped it will not able to convert raw LIDAR and camera data into useful features.

Section B

B1 i) c)

(ii) b)

(iii) a

p

b

c

d

e

(9v) noise & localization affects it give false or fluctuating distance readings and sensor data can't distinguish between the two, and it becomes confused about its location.

B2 (i) a)

(ii) c)

(iii) correct,

1, 3, 4, 6, 2

(9v) Metric map is suitable for the environment, because,

→ It suits for complex areas

→ narrow corridors and escalators

→ It provides dynamic access with heavy human movement.

B3(i) b)

(ii) c)

(iii) Yes, better representation helps to ~~decrease~~ improve localization. It will instead of assuming a fixed position, it maintains a probability distribution. locations.

(iv) 1 d)

2 e) c)

3 ~~a) b)~~ a)

4 b)

5 d) e

Section C

C1 (i) a)

(ii) c)

(iii) C, D, A, B, E, F

C2(i) b)

(ii) b)

(iii)

(iv) It will use dynamic real time updation which is helpful to navigation and in different conditions.

C3

(i) a)

(ii) c)

(iii) Yes DWA is best suitable for real time tracking local path planning and obstacle avoidance technique.

Pr) A* algorithm ~~ess~~ all the path and choose the shortest path for find the goal point, it will check one by one of all the direction to find the output.